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REGULAR PAPER

Motor imagery classification method based on Riemannian Manifold and CNN-LSTM for advanced brain computer interfaces

Yuejiang Hao[1] · Qi Lu[1] · Shiwei Cheng[1]

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Abstract
Motor imagery, a crucial brain–computer interface (BCI) paradigm, provides a new approach to facilitating human–computer interactions. However, its performance is considerably impaired by variances in the operating band and noise in the electroencephalogram (EEG) signal, thus resulting in weak performance. This study proposes a feature extraction method combining a filter bank with Riemannian manifolds for advanced BCIs. The filter bank is used to overcome frequency band variations, and the Riemannian metric is used to reduce noise interference. Covariance matrices are used to construct a high-dimensional Riemannian manifold; they are eventually mapped into feature vectors of the tangent space. Subsequently, a convolutional neural network-long short-term memory (CNN-LSTM) hybrid network is employed to fully utilize the extracted spatial and spectral features (as well as temporal information) for classification. Experiments on the public dataset BCI IV 2a (four classes) and our in-house TMI dataset (three classes) reveal that the proposed method outperforms all existing methods, with classification accuracies of 88.1% and 91.2% for the two datasets, respectively. The proposed method provides a new option for classifying multiclass motor imagery and promises application potential for real-time systems.

Keywords Brain-computer interface · Motor imagery · Deep learning · Riemannian manifold

1 Introduction

Electroencephalogram (EEG) has the advantage of high temporal resolution, making it a favored measure for brain-computer interfaces. An EEG-based BCI is used to translate EEG signals directly into computer commands. Thus, this system enables communication between brain and machines to help those who have motor disabilities (Xie et al. 2022), and has wide potential for applications in neurorehabilitation (Cuomo et al. 2022), gaming (Krepki et al. 2007), and other fields (Yu et al. 2018). Moreover, since EEG can determine human consciousness well, researchers are exploring its application in real-time systems (Kohli et al. 2022). The main limitations of BCIs are their poor usability and lack of robustness. Usability depends on the design of the application, whereas robustness depends on the decoder (Congedo et al. 2017).

✉ Shiwei Cheng
249401866@qq.com

[1] School of Computer Science, Zhejiang University of Technology, Hangzhou 310023, China

Noninvasive BCIs can employ a number of different neural mechanisms. One of the most common active BCI paradigms is motor imagery (MI), wherein users who are handicapped imagine their limbs (such as left hand, right hand, and foot) exhibiting movements to control a system without external stimulation (Zhang et al. 2019). Meanwhile, we cannot ignore the noise brought by the environment. Most of the previous studies demonstrated limited computational efficiency and classification accuracy when integrated into real-time systems. Specifically, low computational efficiency refers to the inability of the algorithms to process data rapidly enough to meet the stringent timing constraints of real-time systems. Additionally, low classification accuracy often results in misinterpretation of the input signals, leading to unreliable or erroneous outputs. These limitations collectively hindered the practical deployment of such methods in real-time applications. Therefore, the solution should be able to capture the key features and react quickly according to the instructions. In this study, we wish to propose a high-performance method for multiclass MI tasks to help BCIs on their path to ubiquity.

A complete BCI system consists of four processes: acquisition of raw signals, preprocessing, feature extraction, and classification. Owing to the non-smoothness and low signal-to-noise ratios (SNRs) of EEG signals, effectively extracting the features of EEG signals and correctly classifying them are the two major challenges for MI–BCIs. Therefore, a more robust feature extraction method can drive the further development of BCIs. In the past decade, numerous common spatial patterns (CSPs) and improved methods have been proposed (Park and Chung 2019; Gubert et al. 2020; Miao et al. 2021) for extracting effective features by constructing optimal spatial filters to distinguish between two classes. CSP-based methods use the arithmetic mean to compute the center of covariance matrices. However, covariance matrices with symmetric positive definite (SPD) forms lie on the Riemannian manifold, whereas EEG signals exist in a curve-shaped space. Consequently, mapping the signals into the defined space is more efficient. Therefore, an increasing number of researchers are exploring Riemannian geometry-based methods (Huang et al. 2021; Miah et al. 2020; Tiwale and Cecotti 2022) to improve the robustness and reduce noise interference. The existing methods have achieved good performance; however, they neglect the selection of the frequency bands and fusion of the features. A filter bank can further improve the performance of CSPs and may also work for Riemannian manifolds. Hence, this study considered combining the two. Herein, mutual information was used to fuse the redundant information brought about by the multiple frequency bands.

Another crucial stage is classification, in previous Riemannian manifold-based methods, traditional machine learning classifiers accounted for the majority of classifiers. However, deep neural networks were rarely used, thus hindering further improvements in classification performance. Deep learning (DL) is often used for learning high-level and latent features from data. DL methods are superior when the data is sufficient and the number of features is large. Thus, numerous researchers have begun to use DL methods for the classification of MI data. Drawing on the concept of DL in speech recognition (Lu et al. 2016), the raw EEG signals are converted into an appropriate intermediate representation without losing the discriminative information of any follow-up DL. The possibility of this method was verified in subsequent work (Chen et al. 2020).

This study proposes a feature extraction method that combines a filter bank with Riemannian manifolds. In particular, first, a raw EEG signal is divided into multiple frequency bands through the filter bank. Next, the Riemannian metric is used to construct the covariance matrices into a high-dimensional Riemannian manifold, the dimensions are reduced by embedding, and suitable features are obtained by operating on the Riemannian tangent space vector. Finally, a convolutional neural network (CNN) combined with a long short-term memory network (LSTM) is designed for classification.

2 Related work

In this section, we presented research advances in three areas: CSP-based methods, Riemannian geometry-based methods, and deep learning classifiers, respectively.

2.1 CSP-based methods

For MI-BCIs, the CSP method is a representative feature extraction method. It extracts effective features by building appropriate spatial filters for distinguishing between two classes. However, the effectiveness of traditional CSP methods relys on the frequency bands selection; moreover, the covariance matrix is susceptible to noise, thus leading to poor cross-subject performance. The subband common spatial pattern (SBCSP) (Novi et al. 2007) and filter bank common spatial pattern (FBCSP) (Ang et al. 2008) have been presented as solutions to the frequency band problem. The SBCSP first divides EEG signals into several subbands and then evaluates the score of each subband for fusion. The FBCSP first separates EEG signals into several bands before extracting features from each band. Finally, the appropriate features are automatically chosen using a feature selection algorithm. Both methods have achieved good results with public datasets.

2.2 Riemannian geometry-based Methods

For the noise problem, a Riemannian mean based on the geometric mean can better describe the central tendency of variance (Congedo et al. 2017). Commonly used Riemannian metrics include the Riemannian mean and tangent space. Correspondingly, two basic methods, namely, the minimum distance to the Riemannian mean (MDRM) and tangent space linear discriminant analysis (TSLDA), have been proposed (Barachant et al. 2011). MDRM uses the Riemannian mean to process the spatial covariance matrix of the EEG directly in the original space. Whereas TSLDA projects the covariance matrix onto the Riemannian tangent space where the matrix can be vectorized and handled as an object of Euclidean geometry; this improves the accuracy, thus resulting in performance that is superior to the CSP method's. Tangent space mapping (TSM) was used to determine the brain activities associated with MI by decomposing an EEG signal into several subbands and estimating the tangent features on each subband (Islam et al. 2018). The TSM allows for the collection of more spatial information by creating higher-dimensional spaces; thus, better classification results are obtained. Although Riemannian geometry-based methods have been proposed for obtaining effective spatial features from MI-EEG signals, the importance of band selection when mapping to Riemannian manifolds has been neglected. If such methods are implemented after a filter bank, the Riemann manifold can preserve the frequency and spatial information of the signal.

2.3 Deep learning

With the development of DL, CNN-based methods have demonstrated excellent performance in EEG signal classification. A single-layer CNN combined with a feature extraction method using CSP and Riemannian geometry had achieved good classification accuracy (Majidov and Whangbo 2019). Although this was a good attempt to integrate Riemannian geometry with DL, it is essentially a binary classification problem. In order to get better approach, a method named CNN-SSD that combined CNN and sparse spectrotemporal decomposition was proposed (Sun et al. 2020). Among them, SSD overcame the shortcomings of traditional time–frequency analysis, and CNN made full use of time–frequency characteristics. The SSD-CNN outperformed traditional classification methods in terms of both kappa values and accuracy. A multi-level feature extraction and fusion model was proposed to construct a robust feature representation of EEG signals using information hidden in different CNN layers (Amin et al. 2019). A mixed-scale CNN architecture using deep convolution and separable convolution was proposed to extract information from different regions at different scales, and classified MI-EEG signals using data enhancement methods (Huang et al. 2020). In general, an EEG signal is a time sequence signal. An LSTM, a special type of recurrent neural network, can fully capture the nonlinear features of time series signals, such as EEG signals. Inspired by natural language processing, combined an LSTM and CNN, and achieved significant improvements on both healthy people and stroke patient datasets (Shen et al. 2017). Furthermore, connecting a CNN and LSTM in parallel instead of using a serial structure provided a new approach for classification studies (Li et al. 2022).

3 Methods

In this section, we presented the technical details of each part of the proposed method, including the spatial covariance matrix, the Riemannian geometry metrics, the construction of Riemannian manifold, Riemannian manifold embedding, Riemannian manifold fusion, and an effective classifier for feature classification.

3.1 Spatial covariance matrix

Different body parts in the MI paradigm correspond to different regions of the motor cortex (such as left hand at the C4 electrode, right hand at C3, and foot at Cz) (Pfurtscheller and Lopes da Silva 1999). Typically, the desired channel is selected and a segment of the EEG signal is split into a number of trials. The i - th trial is denoted as $X_i \in \mathbb{R}^{N \times T_s}$, where N is the number of selected channels and T_s is the time point. When T_s is much larger than N, the spatial covariance matrix can be computed unbiased using the sample covariance matrix (SCM) $P_i \in \mathbb{R}^{N \times N}$, as follows.

$$P_i = \frac{1}{T_s - 1} X_i X_i^T \tag{1}$$

The SCM matrix lies in the space of the SPDs, which are Riemannian manifolds. Therefore, an SPD can benefit from all the mathematical methods described in Riemannian geometry (Xie et al. 2018).

3.2 Riemannian geometry metrics

The Riemannian distance and tangent space are the most crucial and commonly used Riemannian geometric metrics. A curve linking two points on a Riemannian manifold must be at least as long as the Riemannian distance. The Riemannian distance between two SPDs (P_1 and P_2) is defined as follows.

$$\delta_R\left(P_1, P_2\right) = \left\|\log\left(P_1^{-1} P_2\right)\right\|_F = \left[\sum_{i=1}^{n} \log^2 \lambda_i\right]^{\frac{1}{2}} \tag{2}$$

In general, all of the samples in the dataset can be regarded as neighbors, and the mean of all samples, denoted as the Riemannian mean, is regarded as the tangent point P, which can be calculated as follows.

$$P = \underset{P \in SPD(N)}{\arg\min} \sum_{i=1}^{n} \delta_R^2\left(P, P_i\right) \tag{3}$$

It is possible to investigate the nonlinearity of a Riemannian manifold using its tangent space, which is linear (Barachant et al. 2010). The tangent space $T(n)$ at P can be defined as follows.

$$T(n) = \left\{ s_i = upper\left(P^{-\frac{1}{2}} Log_P\left(P_i\right) P^{-\frac{1}{2}}\right) \in \mathbb{R}^{\frac{n(n+1)}{2}} \right\} \tag{4}$$

$$Log_P\left(P_i\right) = P^{\frac{1}{2}} \log\left(P^{-\frac{1}{2}} P_i P^{-\frac{1}{2}}\right) P^{\frac{1}{2}} \tag{5}$$

where $upper(*)$ represents the vectorized upper triangular part of the matrix. The Riemannian distance between P and its neighbor P_i approximates the Euclidean distance in the tangent space, as follows.

$$\delta_R\left(P, P_i\right) \approx \left\|s - s_i\right\|_F \tag{6}$$

3.3 Construction of Riemannian manifold

In order to classify MI-EEG, it is crucial to select the proper band-pass frequency band. Relevant studies pointed out that the α and β bands were the most discriminative for MI tasks including the left and right hands (Pfurtscheller et al. 1997). Therefore, first, the MI period was split in the trial; subsequently, it was filtered with a second-order Butterworth band-pass filter containing six different frequency bands (8–12 Hz, 12–16 Hz, 16–20 Hz, 20–24 Hz, 24–28 Hz, and 28–32 Hz).

All covariance matrices in the same frequency band must be positive definite to apply the Riemannian metrics. For the very few nonpositive definite matrices, a very small value can be added to their diagonals. Subsequently, the Riemannian distance between each pair of matrices was calculated. Further, their adjacencies and weights w_{ij} were determined using k-nearest neighbors, and they were assembled into a high-dimensional Riemannian manifold. The weight between each pair of points P_i and P_j is 0 if they are not adjacent; otherwise, it is calculated as follows (α is a scaling factor).

$$\log_{w_{ij}} = \alpha\delta_R\left(P_i, P_j\right) \tag{7}$$

3.4 Riemannian manifold embedding

Each high-dimensional Riemannian manifold was mapped onto a low-dimensional manifold to preserve the distance structure. At least one neighbor was in common between any two points in the low-dimensional manifold, and the aim of dimensionality reduction was to determine a collection of matrices such that the original close-by points stayed close and the original far-off points stayed far. The mapping with the minimum distance loss was determined as follows (Xie et al. 2016).

$$\min_M \sum_{P_i,P_j \in SPD(N)} \left|\delta_R\left(P_i, P_j\right) - \delta_R\left(MP_iM^T, MP_jM^T\right)\right| \tag{8}$$

3.5 Riemannian manifold fusion

Evidently, the obtained multiple low-dimensional Riemannian manifolds contained considerable redundant information, which could increase the computation time. Therefore, the multiple manifolds must be fused into one while retaining only the most discriminative features.

Given that the points on the low-dimensional Riemannian manifold are in matrix form and cannot be directly merged, TSM was applied to the points before fusion to ensure that the obtained vector retained the original structure to the greatest extent.

Note that mutual information quantifies the correlation between two variables and is independent of the transformations operating on them. Therefore, in this study, the mutual information of each element was computed and the points were fused by preserving the top 25 elements with highest mutual information. The value of each element can be calculated as follows.

$$I_i = H(y) - H(y|V(i,:)) \tag{9}$$

where $H(*)$ represents the entropy calculation (Kumar et al. 2017);y is the label of trial T; and V is the matrix corresponding to a trial T on the low-dimensional Riemannian manifold with one element per row. The illustration is shown in Fig. 1a. In FBRF block, the raw EEG signals are first passed through a filter bank to decompose them into multiple frequency bands, focusing on the alpha and beta bands known to be most discriminative for MI tasks. Each filtered signal is used to compute covariance matrices that are mapped onto a high-dimensional Riemannian manifold. Dimensionality reduction is then applied to map this high-dimensional space into a low-dimensional manifold, preserving the most critical information. Subsequently, mutual information is used to identify and fuse the most discriminative features, retaining only the elements that maximize class separability.

3.6 Proposed convolutional neural network-long short-term memory (CNN-LSTM) network

For the features extracted by the abovementioned method, herein, CNN was combined with an LSTM for classification. The CNN and LSTM can extract the spatial and temporal features, respectively, thereby improving the classification accuracy for the MI-EEG signals. Note that very deep networks can easily lead to overfitting; therefore, this study avoided this when designing the network. As shown in Fig. 1b, three blocks exist with similar structures; however, the size of the kernels in the convolutional layer was sequentially increased to obtain more information. Batch normalization normalized the output of the previous layer and was used in conjunction with a rectified linear unit to effectively reduce the loss and improve the accuracy. A dropout randomly dropped units from the neural network during training, thus preventing overfitting when training with a few samples. Next, two layers of the LSTM were stacked. The first layer set "return_sequences = True" to return all hidden states of the entire sequence to ensure that the second layer could extract deeper features and make the predictions more accurate. Finally, the input was fed to the fully connected layer, and the classification probability was

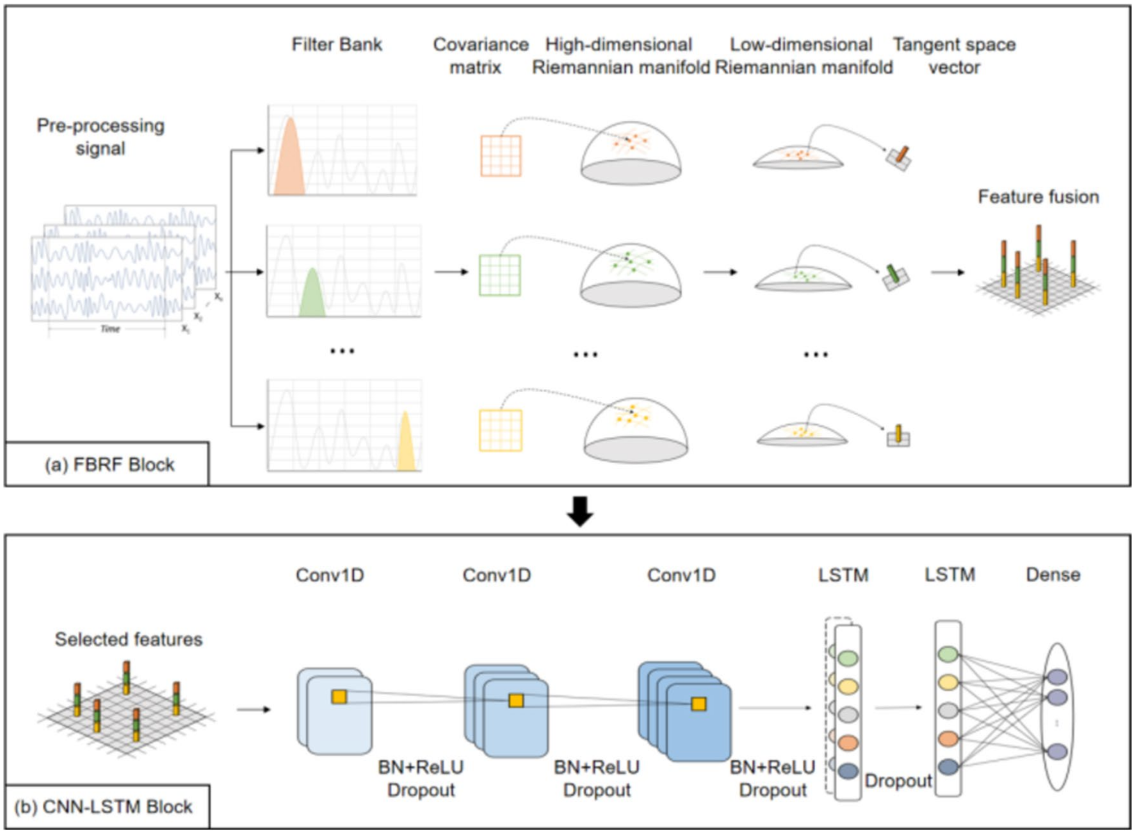

Fig. 1 Schematic of the filter bank Riemannian manifold fusion (FBRF)-convolutional neural network-long short-term memory (CNN-LSTM) architecture. **a** FBRF block for feature extraction and **b** CNN-LSTM block for decoding feature vectors

produced using the softmax. The label with the highest probability was considered the final classification result of the CNN-LSTM. The structure and parameter settings of the network are shown in Table 1.

The CNN-LSTM block is a hybrid architecture that leverages CNN's ability to capture spatial and spectral features and LSTM's strength in modeling temporal dependencies, creating a balanced approach to motor imagery classification. This design carefully avoids the pitfalls of overfitting by using shallow yet effective layers, with batch normalization and dropout ensuring stability during training. By sequentially increasing the kernel sizes in the convolutional layers and stacking two LSTM layers, the model extracts hierarchical representations of the EEG signals, capturing both local and global patterns. This architecture is tailored to handle the complexity of MI-EEG signals, making it highly effective for classifying multi-class motor imagery tasks.

Table 1 Structure and parameter settings of the convolutional neural network-long short-term memory (CNN-LSTM) network

Block	Layer	Filters	Kernel	Strides	Options
1	Conv1D	32	3	2	
	BN				
	ReLU				
	Dropout				$p=0.3$
2	Conv1D	64	3	2	
	BN				
	ReLU				
	Dropout				$p=0.3$
3	Conv1D	128	3	2	
	BN				
	ReLU				
	Dropout				$p=0.3$
4	LSTM	16			return_ sequences = True
	Dropout				$p=0.3$
	LSTM	32			
	Dropout				$p=0.3$
5	Dense	4			Softmax

4 Experiments

Herein, the proposed method was compared with other methods on two datasets, and an ablation study, a sliding window study, and visualization analyses were performed. All experiments were based on MATLAB 9.6, with the biological signal processing tool biosig4octmat 2.80.7, keras 2.5.0 DL framework, and sklearn 0.24.2 machine learning framework. They were run on a computer with a Ryzen 3700X 8-core processor, 16 GB RAM, and an RTX 3080 GPU. The detailed experimental procedure was as follows.

4.1 Data description

(a) The BCI IV 2a dataset (Brunner et al. 2008) included nine subjects (A01–A09), each of whom completed four different MI tasks (left hand, right hand, foot, and tongue). A sampling rate of 250 Hz was used to record the EEG signals. The experimental scheme was as follows. At the beginning ($0-2$ s), a beep sound appeared. After 2 s, a cue of any arrow (to the left, right, down, or up) appeared and lingered on the screen (2–3.25 s). During 3.25–6 s, subjects performed the MI task until the cross disappeared from the screen. Finally, a short rest followed, lasting 1.5 s.

(b) The in-house dataset (TMI) represented a total of 10 subjects (S01–S10) and three types of MI tasks (left hand, right hand, and rest). Each subject knew the experimental procedures and tasks, and signed an informed consent form. All subjects' vision and dominant hands were tested using the Ishihara test (Birch 1997) and Edinburgh Handedness Inventory (Veale 2014). The results revealed that all the subjects were right-handed, and no subjects had any color vision deficits. This study was approved by the authors' university ethics committee. The experiment was performed using a NeuSen W1 wireless digital EEG acquisition system, with an initial sampling rate of 1000 Hz.

Before the experiment began, the subjects were asked to sit in front of a monitor, with an eye-to-screen distance of 50 cm. Note that EEG signals are easily disturbed by the surrounding environment or electromagnetic waves. Hence, a quiet experimental location with a moderate light intensity was chosen to reduce the influences of external factors on the results. Only the subject and researcher were allowed to stay in the experimental room. After the EEG cap was placed on the EEG subject, 64 electrodes were aligned to the corresponding positions of the subject's head according to the 10–20 international system, and a wireless amplifier was added to capture the weak electrical signals. The injection of an EEG paste reduced the impedance between the scalp and electrodes to capture more stable and accurate EEG signals.

Data were collected using the classical MI paradigm. Each trial lasted 8 s. The first 3 s were the rest and preparation times; the monitor displayed a gray cross in the first 2 s to prompt the subjects to get ready, and the cross became bright at 3 s to prompt the subjects that the experiment was about to begin. The left and right arrows or blank pictures appeared on the monitor from 3 to 8 s, and the order was randomized to prevent subjects from forming a memory of the order of the pictures, as this could interfere with the results. In response to the cues, the participants completed the MI tasks. During the experiment, the participants were required to remain relaxed and focused. The dataset consisted of 1500 trials. For each subject, five sessions of experiments were included, each with 30 trials (10 trials for the left hand, right hand, and rest). A 5-min break was inserted between each session of experiments, giving a total experiment duration of approximately 40 min. Figure 2 shows an example of the right-hand experimental paradigm.

4.2 Pre-processing

Before performing feature extraction on the EEG data, the raw MI-EEG signals were preprocessed to improve SNR. Common noises include the 50-Hz power-line interference, electrical signals generated by the subject's eye and head movements, and the DC bias caused by the amplifier of the EEG acquisition device. In this study, the "MNE" toolkit was used for analysis. The EEG signals were first resampled to 250 Hz to reduce the processing time. Next, a high-pass filter (0.5 Hz) and low-pass filter (45 Hz) were used. Subsequently, an average rereferencing was performed to reduce the interference from the individual channel instability. Further, an independent component analysis was performed to remove noise from the eye and head movements. Finally, the

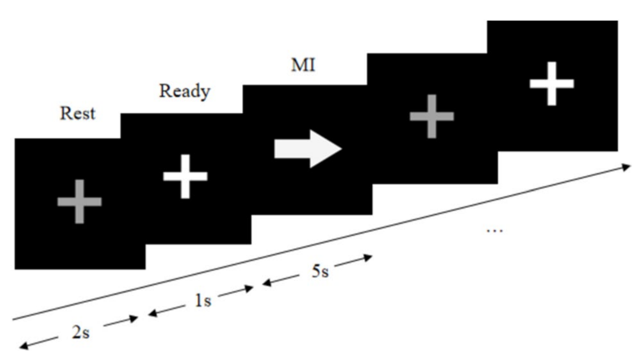

Fig. 2 Right-hand experimental paradigm of TMI dataset

suggested (Saragih et al. 2022) MI-related electrodes (C3, C4, Cz, F3, and F4) were considered.

4.3 Evaluation

The proposed filter bank Riemannian manifold fusion (FBRF)-CNN-LSTM method was evaluated against traditional Riemannian methods (a–c) and deep neural network-based methods (d–f), as follows.

(a) MDRM (Singh et al. 2019): The Riemannian distance between the sample variance matrix and Riemannian mean of each class covariance matrix was calculated, and the minimum distance to the mean was used for classification.
(b) TSLR (Bhattacharyya and Hayashibe 2019): The covariance matrix was mapped to the Riemannian tangent space, where it could be vectorized and handled like a Euclidean object.
(c) MRGF-SVM (Xie et al. 2022): After constructing a Riemannian graph of three frequency bands, bilinear mapping and mutual information were used to extract the features, after then classified by the SVM.
(d) FBSF-TSCNN (Chen et al. 2020): The FBSF returned spatially filtered EEG signals with a temporal representation; subsequently the TSCNN block learnt the discriminated features and returned the decoded results.
(e) DeepConvNet (Schirrmeister et al. 2017): Four convolution-max-pooling blocks were used to perform all computations, all steps were optimized in an end-to-end manner.
(f) EEGNet (Lawhern et al. 2018): The network consists of two blocks, one containing depth-wise convolution and the other containing separable convolution.
(g) PSDTSM-CNN (Majidov and Whangbo 2019): A combination of PSD and TSM is used to extract features, and the particle swarm optimization (PSO) is used to hasten the training, and finally a single-layer CNN is adopted for classification.

4.4 Classification results

Table 2 reports the classification performance on the BCI IV 2a dataset using ten-fold validation. Evidently, the proposed FBRF-CNN-LSTM method outperformed the other methods, with a mean accuracy of 88.1%. The mean accuracy of the FBRF-CNN-LSTM was significantly higher than those of traditional Riemannian geometry-based methods ($p < 0.01$). No significant difference was observed in the accuracy of FBRF-CNN-LSTM compared with PSDTSM-CNN ($p = 0.0875$); However, the p-value was close to 0.05. Thus, the proposed method achieved the highest performance in six out of the nine subjects, more than with any other method. Further, the performances of the FBRF-CNN-LSTM and three other competing Riemannian geometry-based methods were compared based on the TMI dataset. The results are presented in Table 3. Because the TMI dataset was classified for three MI tasks (left hand, right

Table 2 Accuracy of different methods on the test set of BCI IV 2a dataset (the highest mean accuracy is shown in bold)

Method	Subject									Mean Accuracy
	A01	A02	A03	A04	A05	A06	A07	A08	A09	
MDRM	75.3	55.1	69.1	65.5	44.3	46.2	66.9	78.8	72.7	63.8
TSLR	74.9	51.3	72.5	64.8	42.0	48.6	63.9	79.1	78.4	63.9
MRGF-SVM	91.5	59.6	83.2	66.2	44.3	53.0	82.1	78.7	75.9	70.5
EEGNet	83.3	54.1	87.5	63.6	67.4	54.8	88.8	76.8	74.2	72.4
DeepConvNet	84.4	53.1	86.8	62.5	40.3	56.9	86.8	77.8	80.6	69.9
FBSF-TSCNN	85.8	60.1	87.8	64.2	48.6	56.9	83.0	81.6	80.2	72.0
PSDTSM-CNN	90.2	76.4	91.2	**84.4**	**76.6**	**74.8**	92.4	91.6	92.5	85.3
FBRF-CNN-LSTM (ours)	**93.4**	**86.2**	**98.4**	82.6	72.7	72.4	**97.4**	**95.5**	**94.6**	**88.1**

Table 3 Accuracy of Riemannian geometry-based methods on TMI dataset (the highest mean accuracy is shown in bold)

Method	Subject										Mean Accuracy
	S01	S02	S03	S04	S05	S06	S07	S08	S09	S10	
MDRM	74.7	84.0	64.4	71.3	80.3	66.4	68.5	70.9	80.2	74.7	73.5
TSLR	78.2	84.4	66.8	76.8	81.2	67.3	67.6	76.1	80.1	78.5	75.6
MRGF-SVM	83.8	90.7	71.8	83.3	90.2	75.3	80.5	82.3	86.6	85.0	82.9
FBRF-CNN-LSTM (ours)	**93.1**	**99.4**	**83.1**	**91.8**	**96.3**	**84.2**	**84.8**	**91.1**	**94.7**	**93.6**	**91.2**

hand, and rest), for simplicity, accuracy was used as the performance metric. Notably, the advantage of the FBRF-CNN-LSTM was further enhanced by the high quality of the acquired EEG signals owing to the fewer interference sources in the laboratory. The mean accuracies of the proposed method were 17.8%, 15.7%, and 8.4% higher than those of the MDRM, TSLR, and MRGF-SVM approaches, respectively ($p < 0.01$). In addition, the proposed method achieved the highest performance among the ten subjects, demonstrating its superior classification ability.

4.5 Ablation study

The proposed model was further evaluated through an ablation study, wherein each component was iteratively removed and the model was retrained to predict the data with a test set to study their impacts. The results from the ablation study performed using the TMI dataset are presented in Table 4. Evidently, removing any component decreased the accuracy. Removing block2 decreased the classification accuracy the most, which implies that this component captures the most important features. Furthermore, when the FBRF was removed, the standard deviation (Std) was higher than that of the proposed method, thus indicating that the FBRF method was able to extract common features across subjects. These results show that the proposed method can considerably enhance the classification performance.

Table 4 Ablation study on the TMI dataset (the highest mean accuracy and Std are shown in bold)

Method	Mean Accuracy(%)	Std
w/o FBRF	87.5	7.2
w/o Block1	82.6	6.7
w/o Block2	84.9	5.9
w/o Block3	85.2	5.6
w/o Block4	84.3	5.8
Full (proposed)	**91.2**	**5.2**

4.6 Separability visualization

To provide more intuitive results, the t-distributed stochastic neighbor embedding method (Maaten and Hinton 2008) was used for dimensionality reduction and visualization, thereby converting the features of the BCI IV 2a and TMI datasets into 2D representations. As shown in Fig. 3, (a)–(c) are the BCI IV 2a datasets; red, orange, green, and purple denote the left hand, right hand, feet, and tongue, respectively; and (d)–(f) are the TMI datasets; red, yellow, and purple denote the left hand, right hand, and rest, respectively. Each column represents the distribution of the raw signals after the FBRF and CNN-LSTM. Evidently, the raw signals of different classes were difficult to distinguish, and the distance between different classes gradually increased after being processed by the proposed method. These results provide evidence of the high classification performance of the proposed method.

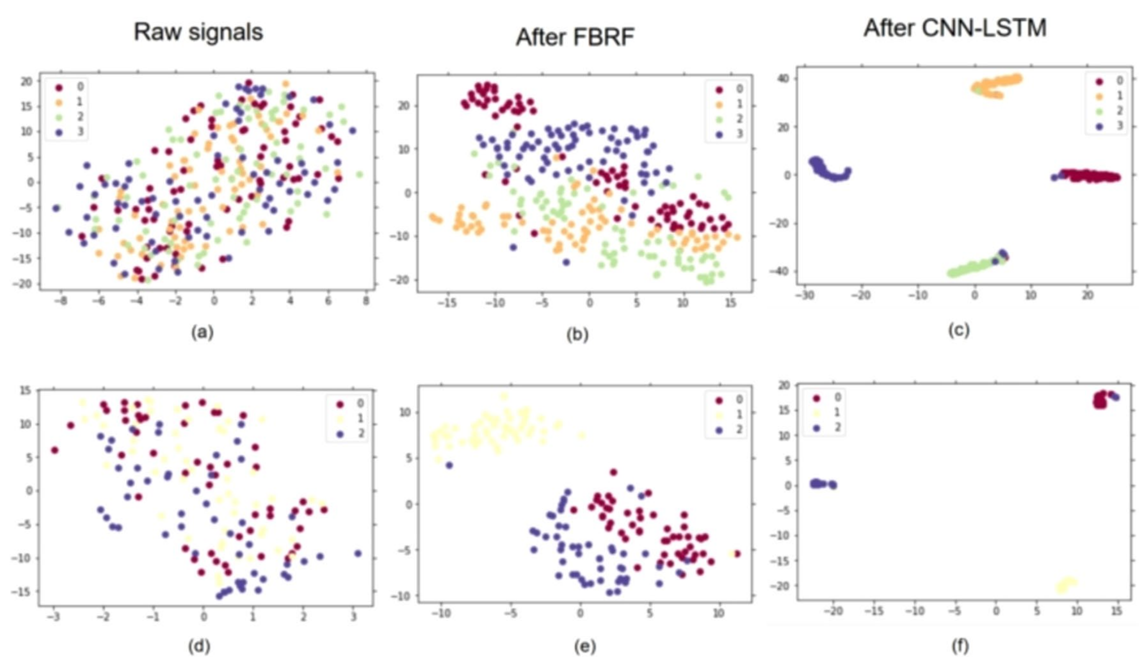

Fig. 3 t-distributed stochastic neighbor embedding (*t*-SNE) visualization of A01 and S01 in two datasets

5 Discussion

In this study, we presented an effective method for classifying MI-EEG signals and evaluated it on two datasets. The majority achieved an accuracy considerably above chance level, and some subjects even reached 99.4%. On the other hand, there were a few subjects who fell below expectations. There are several possible explanations for the results. Although we carried out data cleansing in the pre-processing stage, the interference of artifacts cannot be completely avoided. For subjects with lower classification accuracy, it is possible that their data were of lower quality. In this section, we focused on discussing why this method works, the limitations encountered so far, and potential directions for future research.

5.1 Separability visualization

Inspired by the FBCSP, feature extraction and selection from subbands was adopted when designing the FBRF block to solve the variance in the operational frequency band. The FBCSP has finer frequency bands (nine filters, ranging from 4 to 40 Hz), whereas the FBRF designs six filters for the alpha and beta frequency bands, i.e., the most discriminative for MI tasks. Second, the feature selection in the FBCSP is based on the mutual information of the CSP, whereas the FBRF assembles the covariance matrices into a high-dimensional Riemannian manifold and performs embedding for dimensionality reduction, which is eventually vectorized. The traditional CSP is based on Euclidean geometry and uses the arithmetic mean to estimate the covariance matrix; however, the arithmetic mean cannot guarantee the invariance of the matrix inversion, and the swelling effect brought by leads to distortion. Because the covariance matrix is SPD in construction, it is essentially a Riemannian manifold. Therefore, methods based on Riemannian geometry can better capture the underlying geometry and reduce noise interference.

5.2 Method effectiveness

The proposed method achieves superior performance in multiclass motor imagery classification by addressing key limitations in traditional methods. Specifically:

The FBRF block is designed to address two critical challenges in EEG-based BCI systems: frequency band variations and noise interference. By decomposing raw EEG signals into multiple frequency bands, the FBRF block ensures that the most relevant alpha and beta bands—known to be highly discriminative for MI tasks—are preserved. Furthermore, the use of covariance matrices mapped to a Riemannian manifold allows for more effective representation of EEG signal features while minimizing the impact of noise. Dimensionality reduction and mutual information-based feature fusion further ensure that the extracted features are not only robust but also computationally efficient.

On the other hand, the hybrid CNN-LSTM block capitalizes on the strengths of both architectures. The CNN component extracts spatial and spectral features from the processed EEG data, while the LSTM component models temporal dependencies, capturing the dynamic nature of MI tasks. This complementary structure ensures a more holistic representation of the EEG signals, leading to improved classification performance.

The proposed method's effectiveness is further supported by experimental results. Achieving a mean accuracy of 88.1% on BCI IV 2a dataset and a mean accuracy of 91.2% on TMI dataset.

5.3 Limitations and future work

Although the proposed method achieved good performance in MI classification, the current study still suffers from limitations, which will motivate our future research. First, only five channels were used in our methods, and more channels mean better usability of BCIs and higher accuracy. However, as the dimension of the covariance matrix increases with more channels, the computational efficiency of Riemannian geometry-based methods decreases. To achieve a balance between usability and accuracy, channel selection can be optimized using techniques such as mutual information-based selection and recursive feature elimination (RFE). Mutual information-based selection evaluates the relevance of each EEG channel by measuring its mutual information with the classification labels, thereby identifying the most discriminative channels while reducing redundancy. Besides, recursive feature elimination iteratively removes the least significant channels by assessing their contribution to classification accuracy, ultimately retaining only the most informative subset of channels. Second, EEG is known to change with the individual's mental state, surrounding environment, and other factors. Thus, the pre-trained model should be able to update automatically to ensure that the accuracy is maintained at a satisfactory level. An additional idea, referring to the use of a feature pyramid in the field of computer vision to generate regions at different scales for target detection (Lin et al. 2017), signal segments from different time windows can be intercepted and stacked into the EEG pyramid in the future.

6 Conclusion

This study proposed a novel FBRF feature extraction method that combines a filter bank and Riemannian manifold. This method effectively extracts features from EEG signals, and a CNN-LSTM neural network is designed for classification. The proposed method was evaluated on the BCI IV 2a public dataset and TMI dataset. The results revealed that the proposed staged method performed better than several traditional Riemannian geometry-based methods, as well as end-to-end DL methods. Furthermore, an ablation study was conducted to show the impact of each part of our method on performance. The proposed method achieved cutting-edge performance and superior separability. Thus, it provides a new approach to MI-EEG classification and shows promising application potential for real-time MI-BCIs.

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Data availability The public dataset BCI IV 2a can be found on https://www.bbci.de/competition/iv, and the inhouse TMI dataset is not publicly available but is available on reasonable request.

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

References

Amin, S.U., Alsulaiman, M., Muhammad, G., Bencherif, M.A., Hossain, M.S.: Multilevel weighted feature fusion using convolutional neural networks for EEG motor imagery classification. IEEE Access **7**, 18940–18950 (2019)

Ang, K.K., Chin, Z.Y., Zhang, H., Guan, C.: Filter bank common spatial pattern (FBCSP) in brain-computer interface. In: Proceedings of 2008 IEEE International Joint Conference on Neural Networks (IEEE World Congress on Computational Intelligence). IEEE, Hong Kong, pp. 2390–2397 (2008)

Barachant, A., Bonnet, S., Congedo, M., Jutten, C.: Multiclass brain–computer interface classification by Riemannian geometry. IEEE Trans. Biomed. Eng. **59**(4), 920–928 (2011)

Barachant, A., Bonnet, S., Congedo, M., Jutten, C.: Riemannian geometry applied to BCI classification. In: Proceedings of LVA/ICA. Springer, Berlin, Heidelberg, pp. 629–636 (2010)

Bhattacharyya, S., Hayashibe, M.: Systematic enhancement of functional connectivity in brain-computer interfacing using common spatial patterns and tangent space mapping. arXiv preprint arXiv:1907.08977 (2019)

Birch, J.: Efficiency of the Ishihara test for identifying red-green colour deficiency. Ophthalm. Physiol. Opt. **17**(5), 403–408 (1997)

Brunner, C., Leeb, R., Müller-Putz, G., Schlögl, A., Pfurtscheller, G.: BCI Competition 2008–Graz data set A. In: Institute for Knowledge Discovery (Laboratory of Brain-Computer Interfaces), Graz University of Technology, pp. 1–6 (2008)

Chen, J., Yu, Z., Gu, Z., Li, Y.: Deep temporal-spatial feature learning for motor imagery-based brain–computer interfaces. IEEE Trans. Neural Syst. Rehabil. Eng. **28**(11), 2356–2366 (2020)

Congedo, M., Barachant, A., Bhatia, R.: Riemannian geometry for EEG-based brain-computer interfaces; a primer and a review. Brain-Comput. Interfaces **4**(3), 155–174 (2017)

Cuomo, G., Maglianella, V., Ghooshchy, S.G., Zoccolotti, P., Martelli, M., Paolucci, S., Morone, G., Iosa, M.: Motor imagery and gait control in Parkinson's disease: techniques and new perspectives in neurorehabilitation. Expert Rev. Neurother. **22**(1), 43–51 (2022)

Gubert, P.H., Costa, M.H., Silva, C.D., Trofino-Neto, A.: The performance impact of data augmentation in CSP-based motor-imagery systems for BCI applications. Biomed. Signal Process. Control **62**, 102152 (2020)

Huang, W., Wang, L., Yan, Z., Liu, Y.: Classify motor imagery by a novel CNN with data augmentation. In: Proceedings of 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC). IEEE, Montreal, pp. 192–195 (2020)

Huang, S., Cai, G., Wang, T., Ma, T.: Amplitude-phase information measurement on Riemannian manifold for motor imagery-based BCI. IEEE Signal Process. Lett. **28**, 1310–1314 (2021)

Islam, M.R., Tanaka, T., Molla, M.K.I.: Multiband tangent space mapping and feature selection for classification of EEG during motor imagery. J. Neural Eng. **15**(4), 046021 (2018)

Kohli, V., Tripathi, U., Chamola, V., Rout, B.K., Kanhere, S.S.: A review on virtual reality and augmented reality use-cases of brain computer interface based applications for smart cities. Microprocess. Microsyst. **88**, 104392 (2022)

Krepki, R., Blankertz, B., Curio, G., Müller, K.-R.: The Berlin Brain-Computer Interface (BBCI)–towards a new communication channel for online control in gaming applications. Multimedia Tools Appl. **33**(1), 73–90 (2007)

Kumar, S., Sharma, A., Tsunoda, T.: An improved discriminative filter bank selection approach for motor imagery EEG signal classification using mutual information. BMC Bioinform. **18**, 125–137 (2017)

Lawhern, V.J., Solon, A.J., Waytowich, N.R., Gordon, S.M., Hung, C.P., Lance, B.J.: EEGNet: a compact convolutional neural network for EEG-based brain–computer interfaces. J. Neural Eng. **15**(5), 056013 (2018)

Li, H., Ding, M., Zhang, R., Xiu, C.: Motor imagery EEG classification algorithm based on CNN-LSTM feature fusion network. Biomed. Signal Process. Control **72**, 103342 (2022)

Lin, T.-Y., Dollar, P., Girshick, R., He, K., Hariharan, B., Belongi, S.: Feature pyramid networks for object detection. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Honolulu, Hawaii, pp. 2117–2125 (2017)

Lu, N., Li, T., Ren, X., Miao, H.: A deep learning scheme for motor imagery classification based on restricted Boltzmann machines. IEEE Trans. Neural Syst. Rehabil. Eng. **25**(6), 566–576 (2016)

Majidov, I., Whangbo, T.: Efficient classification of motor imagery electroencephalography signals using deep learning methods. Sensors **19**(7), 1736 (2019)

Miah, A.S.M., Rahim, M.A., Shin, J.: Motor-imagery classification using riemannian geometry with median absolute deviation. Electronics **9**(10), 1584 (2020)

Miao, Y., Jin, J., Daly, I., Zuo, C., Wang, X., Cichocki, A., Jung, T.-P.: Learning common time-frequency-spatial patterns for motor imagery classification. IEEE Trans. Neural Syst. Rehabil. Eng. **29**, 699–707 (2021)

Novi, Q., Guan, C., Dat, T.H., Xue, P.: Sub-band common spatial pattern (SBCSP) for brain-computer interface. In Proceedings of the 3rd International IEEE/EMBS Conference on Neural Engineering. IEEE, Kohala Coast, pp. 204–207 (2007)

Park, Y., Chung, W.: Frequency-optimized local region common spatial pattern approach for motor imagery classification. IEEE Trans. Neural Syst. Rehabil. Eng. **27**(7), 1378–1388 (2019)

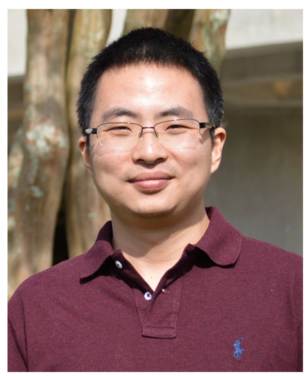
Pfurtscheller, G., Lopes da Silva, F.H.: Event-related EEG/MEG synchronization and desynchronization: basic principles. Clin. Neurophysiol. **110**(11), 1842–1857 (1999)

Pfurtscheller, G., Neuper, C., Flotzinger, D., Pregenzer, M.: EEG-based discrimination between imagination of right and left hand

movement. Electroencephalogr. Clin. Neurophysiol. **103**(6), 642–651 (1997)
Saragih, A.S., Basyiri, H.N., Raihan, M.Y.: Analysis of motor imagery data from EEG device to move prosthetic hands by using deep learning classification. In: Proceedings of AIP Conference. AIP, Melville, pp. 050009 (2022)
Schirrmeister, R.T., Springenberg, J.T., Fiederer, L.D.J., Glasstetter, M., Eggensperger, K., Tangermann, M., Hutter, F., Burgard, W., Ball, T.: Deep learning with convolutional neural networks for EEG decoding and visualization. Hum. Brain Mapp. **38**(11), 5391–5420 (2017)
Shen, Y., Lu, H., Jia, J.: Classification of motor imagery EEG signals with deep learning models. In: Proceedings of Intelligence Science and Big Data Engineering: 7th International Conference, IScIDE, pp. 181–190. Springer, Dalian (2017)
Singh, A., Lal, S., Guesgen, H.W.: Small sample motor imagery classification using regularized Riemannian features. IEEE Access **7**, 46858–46869 (2019)
Sun, B., Zhao, X., Zhang, H., Bai, R., Li, T.: EEG motor imagery classification with sparse spectrotemporal decomposition and deep learning. IEEE Trans. Autom. Sci. Eng. **18**(2), 541–551 (2020)
Tiwale, G., Cecotti, H.: Motor imagery classification using riemannian geometry in multiple frequency bands with a weighted nearest neighbors approach. In: Proceedings of the Pattern Recognition: 14th Mexican Conference, MCPR, pp. 159–168. Springer, Mexico (2022)
van der Maaten, L., Hinton, G.: Visualizing data using t-SNE. J. Mach. Learn. Res. **9**, 11 (2008)
Veale, J.F.: Edinburgh handedness inventory–short form: a revised version based on confirmatory factor analysis. Laterality: Asymmetr. Body Brain Cogn. **19**(2), 164–177 (2014)
Xie, X., Liang, Z., Yu, H., Lu, Z.G., Li, Y.: Motor imagery classification based on bilinear sub-manifold learning of symmetric positive-definite matrices. IEEE Trans. Neural Syst. Rehabil. Eng. **25**(6), 504–516 (2016)
Xie, X., Yu, Z.L., Gu, Z., Zhang, J., Cen, L., Li, Y.: Bilinear regularized locality preserving learning on Riemannian graph for motor imagery BCI. IEEE Trans. Neural Syst. Rehabil. Eng. **26**(3), 698–708 (2018)
Xie, X., Zou, X., Yu, T., Tang, R., Hou, Y., Qi, F.: Multiple graph fusion based on Riemannian geometry for motor imagery classification. Appl. Intell. **52**(8), 9067–9079 (2022)
Yu, Y., Liu, Y., Jiang, J., Yin, E., Zhou, Z., Hu, D.: An asynchronous control paradigm based on sequential motor imagery and its application in wheelchair navigation. IEEE Trans. Neural Syst. Rehabil. Eng. **26**(12), 2367–2375 (2018)
Zhang, R., Zong, Q., Dou, L., Zhao, X.: A novel hybrid deep learning scheme for four-class motor imagery classification. J. Neural Eng. **16**(6), 066004 (2019)

Yuejiang Hao received the B.S. degree in Software Engineering from Zhejiang University of Technology, Hangzhou, China, in 2022. He is currently pursuing the M.S. degree in Software Engineering at the School of Software Engineering, Zhejiang University of Technology, from 2022 to 2025. His research interests include EEG-based motor imagery classifacation and machine learning.

Qi Lu received the B.S. degree in Computer Science and Technology from Jiaxing University, JiaXing, China, in 2020, and the M.S. degree in Computer Technology from Zhejiang University of Technology, HangZhou, China, in 2023. His research interests include biomedical signal processing and machine learning.



Shiwei Cheng received the Ph.D. degree in computer science and technology from Zhejiang University, Hangzhou, China in 2009. In 2009, he joined the School of Computer Science and Technology at Zhejiang University of Technology, where he became a Professor. He served as committee member of SIGCHI China Chapter, China Computer Federation (CCF) HCI Technical Committee. His research interests include human-computer interaction, ubiquitous computing and collaborative computing.